

9 Ans: B

Lorry is accelerated from rest to a speed of 100 km h^{-1} .

Work done by lorry's engine = increase in lorry's K.E.

Friction has to be ignored as minimum time is to be considered

$$Pt = \frac{1}{2}mv^2$$

$$t = \frac{mv^2}{2P}$$

where $v = 100 \text{ km h}^{-1} = 27.78 \text{ m s}^{-1}$

$$\therefore t = \frac{(2000)27.78^2}{2 \times 50000} = 15.4 \text{ s}$$

10 Ans: D